

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
SSCS, Midterm Examination Spring 2026

Time: 90 minutes

Course Title (CT-261) – B

Max Marks: 20

Q1) Discuss view. Why are they important in a database approach?

[CLO-1]

Miss Amna [4 Marks]

Q2) Discuss three levels of abstraction in ANSI-SPARC architecture of DBMS.

[CLO-1]

Dr Umer [6 Marks]

Q3) Discuss each of the following concepts in the context of the relational data model:

- a) Relation
- b) Domain
- c) Degree and Cardinality

[CLO-2]

[3 Marks]

Q4) Suppose you set up a database company, ArtBase, that builds a product for art galleries. The core of this product is a database with a schema that captures all the information that galleries need to maintain. Galleries keep information about artists, their names (which are unique), birthplaces, age, and style of art. For each piece of artwork, the artist, the year it was made, its unique title, its type of art (e.g., painting, lithograph, sculpture, photograph), and its price must be stored. Pieces of artwork are also classified into groups of various kinds, for example, portraits, still lifes, works by Picasso, or works of the 19th century; a given piece may belong to more than one group. Each group is identified by a name (like those just given) that describes the group. Finally, galleries keep information about customers. For each customer, galleries keep that person's unique name, address, total amount of dollars spent in the gallery (very important!), and the artists and groups of art that the customer tends to like.

Design an ER diagram to capture the above requirements. State any assumptions you have that affect your design. Make sure cardinalities and primary keys are clear.

[CLO-2]

[7 Marks]